Descriptive Epidemiology

What Is Descriptive Epidemiology?

- **Purpose:** Describe how health and disease are distributed across populations
- **Core Questions**
  - **Who** is affected? *(Person)*
  - **Where** does it occur? *(Place)*
  - **When** does it occur? *(Time)*
- **Why It Matters**
  - Monitor health and disease trends
  - Guide health services planning
  - Identify priorities for further study

Example: Descriptive Epidemiology in Practice

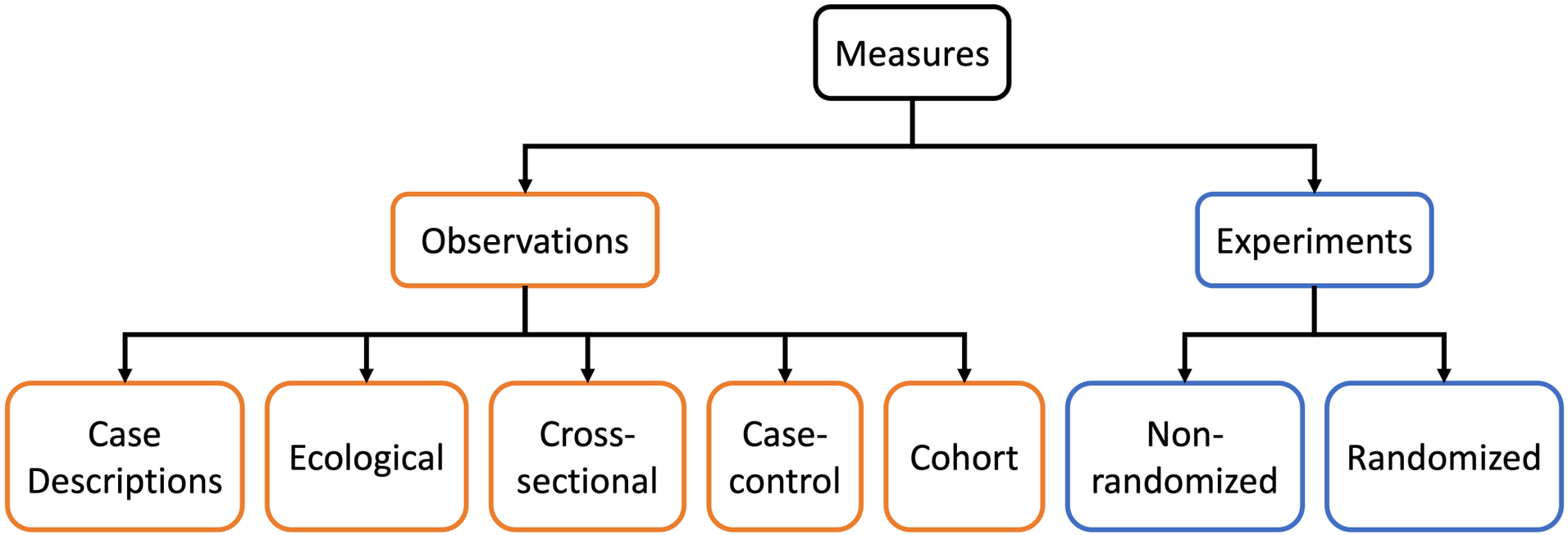
- **Scenario:** Breastfeeding trends among U.S. infants born in 2012:
  - **43%** exclusively breastfed for the first **3 months**
  - **22%** exclusively breastfed for **6 months**
- **What This Describes**
  - **Person:** U.S. infants
  - **Place:** United States
  - **Time:** First 3 vs. 6 months after birth
- **Discussion Questions**
  - What **patterns** do you observe across time?
  - What **additional descriptive data** would help clarify this pattern (e.g., age, SES, region)?
  - Which **groups might be most affected** by this decline?

Descriptive Epidemiologic Study Designs

Descriptive Epidemiologic Study Designs

- **This section focuses on three common descriptive designs:**
  - **Case reports**
  - **Case series**
  - **Cross-sectional studies**
  - Other descriptive designs exist (e.g., **ecologic studies**, **surveillance-based descriptions**)
- **What these designs have in common**
  - Describe **who**, **where**, and **when**
  - Do **not** test causal relationships
  - Often provide the **first signal** of a health issue or disparity

Hierarchy of Study Designs



Typically less convincing → Typically more convincing

Case Reports

- **Definition:** A detailed description of a **single patient** or a **small number of unusual cases**, typically reported **individually**, not as a group analysis.
- **Purpose**
  - Document **novel, rare, or unexpected** health events
  - Alert clinicians and public health officials to **new or emerging issues**
- **Why They Matter**
  - Often provide the **first signal** of a new disease, complication, or side effect
  - Contribute to early **public health awareness**
  - Serve as educational examples in clinical and public health settings
- **Key Limitation**
  - No comparison group → **cannot estimate risk or prevalence**

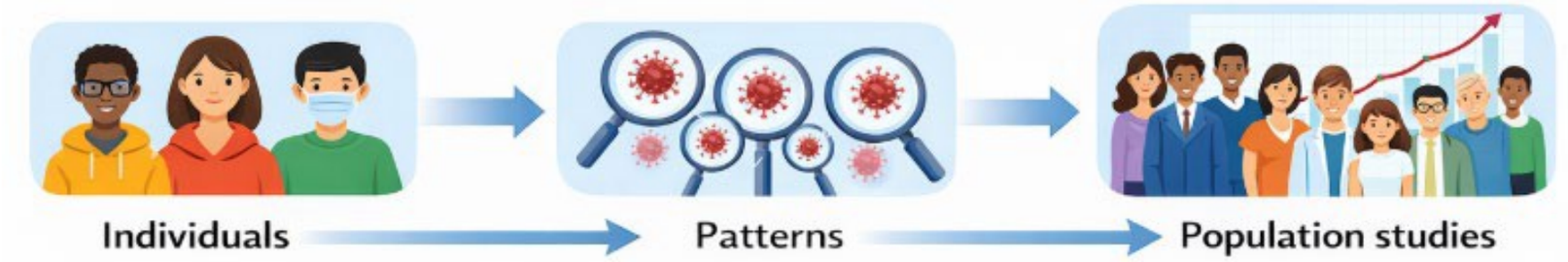
Case Series

- **Definition:** A descriptive study that reports on **multiple patients** with the **same diagnosis or condition**, described **collectively**.
  - **Case report:** Each case is described **on its own**
  - **Case series:** Multiple cases are described **together to identify patterns**
- **What Case Series Can Describe**
  - Common **clinical features**
  - Typical **disease progression**
  - Distribution by **person, place, or time**
- **Key Limitation:** Cannot estimate **risk**, **rates**, or **causal effects**

Why Individual Cases Belong in Epidemiology

“Isn’t epidemiology about populations?”

- **Yes — but populations are built from individual cases.**
- **Case reports and case series** focus on individuals → to **identify and describe new or unusual health events**
- These early descriptions:
  - Reveal **what counts as a case**
  - Signal **emerging health problems**
  - Motivate **population-level investigation**

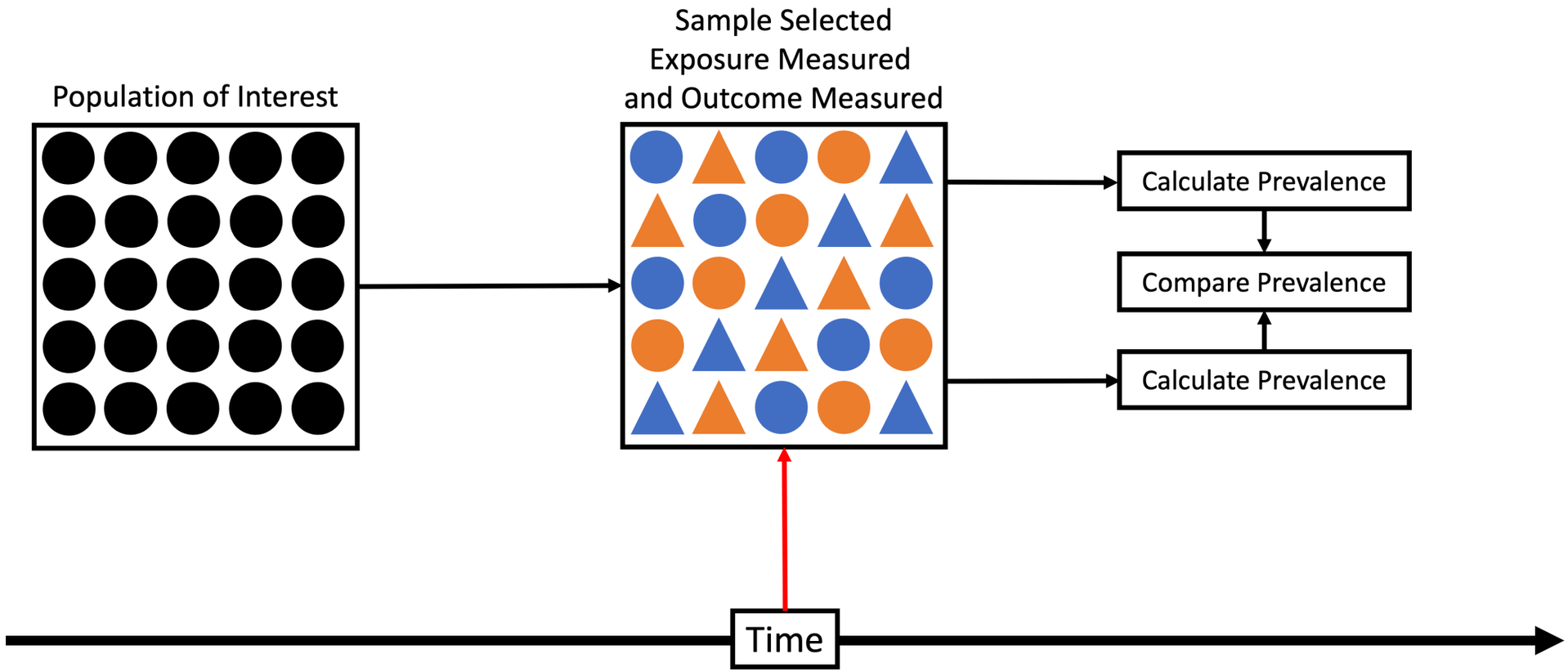


Cross-Sectional Studies

- **Definition:** A descriptive study that examines **health outcomes and related characteristics** across a defined **population at a single point in time**.
- **Key Features**
  - Data collected from **individuals**
  - Results describe **population-level patterns**
  - Measures **prevalence**, not incidence
  - Exposure(s) and outcome(s) assessed **simultaneously**
- **Core Focus: A snapshot** describing how health outcomes are distributed **across a population**

Cross-Sectional: One-Time Snapshot

Basic cross-sectional study design



Project Prep 2

Person, Place, and Time

Person Variables: The “Who” Factor

- **Definition:** Person variables are demographic characteristics that can influence health outcomes.
- **Purpose:** Understanding who is at risk helps tailor public health interventions.
- **Key Person Variables:**
  - Age, Sex, Race/ethnicity, Socioeconomic status
  - Other examples: Gender, Marital status, Nativity (place of origin), Religion

Age

- **Definition:** Age describes how disease occurrence and health outcomes vary across the life course.
- **Why Age Matters**
  - Disease patterns often differ by **life stage**
  - Helps identify **age-specific health priorities**
- **Examples**
  - **Childhood:** Higher occurrence of some infectious diseases
  - **Young adults:** Higher rates of injury-related outcomes
  - **Older adults:** Higher prevalence of chronic disease and mortality
- **Discussion Question:** What health concerns are most common among **college-aged adults**?

Sex/Gender

- **Definition:** Sex and gender describe differences in health outcomes related to **biological characteristics** and **social roles, behaviors, and norms**.
- **Why Sex/Gender Matters**
  - Many diseases show different **patterns of occurrence and outcomes** by sex or gender
  - Differences may reflect **biology, behavior, social context, or access to care**
- **Examples**
  - **Heart disease:** Higher rates and earlier onset among males
  - **Autoimmune diseases & depression:** Higher prevalence among females
  - **Mortality:** All-cause age-specific mortality rates are typically higher among males
- **Discussion Question:** What health concerns or health-related behaviors differ by **sex or gender** among college students?

Race/Ethnicity

- **Definition:** Race and ethnicity describe population groups based on **social, cultural, historical, and geographic characteristics**.
- **Why Race/Ethnicity Matters**
  - Disease occurrence and health outcomes often vary across groups
  - Patterns reflect **inequities in exposure, resources, and access to care**, not biology alone
- **Examples of Observed Patterns**
  - **Asthma:** Lower reported prevalence among some Hispanic populations
  - **Cognitive decline:** Higher prevalence among American Indian or Alaska Native populations
  - **COVID-19:** Higher mortality rates among non-Hispanic Black populations, especially early in the pandemic
- **Discussion Question:** What factors—other than biology—might help explain differences in health outcomes across racial or ethnic groups?

Socioeconomic Status (SES)

- **Definition:** Socioeconomic status reflects an individual's or group's **economic and social position**, commonly measured by **income, education, and occupation**.
- **Why SES Matters**
  - Strongly associated with differences in **health outcomes and access to care**
  - Patterns often show a **social gradient**, with worse health at lower SES levels
- **Examples of SES-Related Patterns**
  - Higher rates of chronic disease and mortality in lower-SES groups
  - Reduced access to preventive services and timely care
  - Greater exposure to environmental and occupational risks
- **Discussion Question:** Which aspects of SES might most strongly influence health among **college students**?

Social Determinants of Health (SDOH)

- **Definition:** Social determinants of health are the **conditions in which people are born, grow, live, work, and age** that influence health outcomes.
- **Why SDOH Matter**
  - Shape **opportunities for health** beyond individual behaviors
  - Help explain **persistent health disparities** across populations
- **Common SDOH Domains**
  - Economic stability: e.g., ability to afford food, housing, or health care
  - Education access and quality: e.g., access to quality schools or educational resources
  - Health care access and quality: e.g., availability of primary care or mental health services
  - Neighborhood and built environment: e.g., safe housing and access to healthy foods
  - Social and community context: e.g., social support or community connectedness
- **Discussion Question:** Which SDOH are most relevant to health outcomes among **college students**?

SES vs. SDOH

	Socioeconomic Status (SES)	Social Determinants of Health (SDOH)
Concept	Social position	Living and working conditions
What it describes	Social and economic standing	Conditions shaping daily life and health
Common measures / examples	Income, education, occupation	Housing, working conditions, access to care, education, social support

Exercise: SES or SDOH?

For each example, decide whether it best represents **SES** or **SDOH**. Reminder: SES reflects **position**; SDOH reflect **conditions that come with that position**.

- A person's highest level of education
- Living in a neighborhood with limited access to grocery stores
- Type of job and employment status
- Exposure to hazardous conditions at work
- Ability to take paid sick leave
- Household income level

Health Disparities

- **Definition:** Health disparities are **differences in health outcomes, disease burden, or access to care** across population groups.
- **Why Health Disparities Matter**
  - Indicate **inequities** in health and health care
  - Reflect differences in **SES, SDOH, and social context**
- **Examples of Health Disparities**
  - Higher rates of chronic disease in lower-SES populations
  - Differences in access to preventive care by income or insurance status
  - Higher mortality rates for certain conditions across racial or ethnic groups
- **Key Point:** Health disparities are **descriptive patterns**, not explanations.
- **Discussion Question:**
  - Which person-related factors (age, race/ethnicity, SES, SDOH) are most likely contributing to health disparities among **college students**?

Place Variables: The “Where” Factor

- **Definition:** Place describes how disease occurrence and health outcomes vary by **geographic location**.
- **Why Place Matters**
  - Health patterns often differ across locations due to **environmental, social, and structural factors**
  - Helps identify **geographic disparities** in health and access to care
- **Common Place Levels**
  - International
  - National (within-country)
  - Regional or state
  - Urban vs. rural
  - Local or neighborhood

International Variation

- **Definition:** Differences in disease patterns and health outcomes **across countries**.
- **Why Place Matters Internationally**
  - Health outcomes vary widely due to **contextual differences**
  - Reflect differences in **health systems, environments, and resources**
- **Examples**
  - **Infectious diseases:** Polio remains endemic in a few countries while eliminated elsewhere
  - **Life expectancy:** Substantial differences across countries and regions
  - **Maternal and child health:** Higher mortality in lower-resource settings
- **Key Point:** International variation reflects **where people live**, not individual risk alone.
- **Discussion Question**
  - What factors might contribute to health differences between countries?

National (Within-Country) Variations

- **Definition:** Differences in disease patterns and health outcomes **across regions or locations within the same country**.
- **Why Place Matters Within Countries**
  - Health outcomes vary by **region, state, and community**
  - Reflect differences in **environment, resources, policies, and access to care**
- **Examples**
  - **Heart disease:** Rates vary substantially by U.S. state and region
  - **Respiratory illness:** Higher prevalence in areas with greater air pollution
  - **Injury and violence:** Higher rates in some regions compared with others
- **Key Point:** Health patterns can differ greatly **even within the same country**.
- **Discussion Question**
  - What factors might explain health differences between regions within the U.S.?

National Variation (Example)

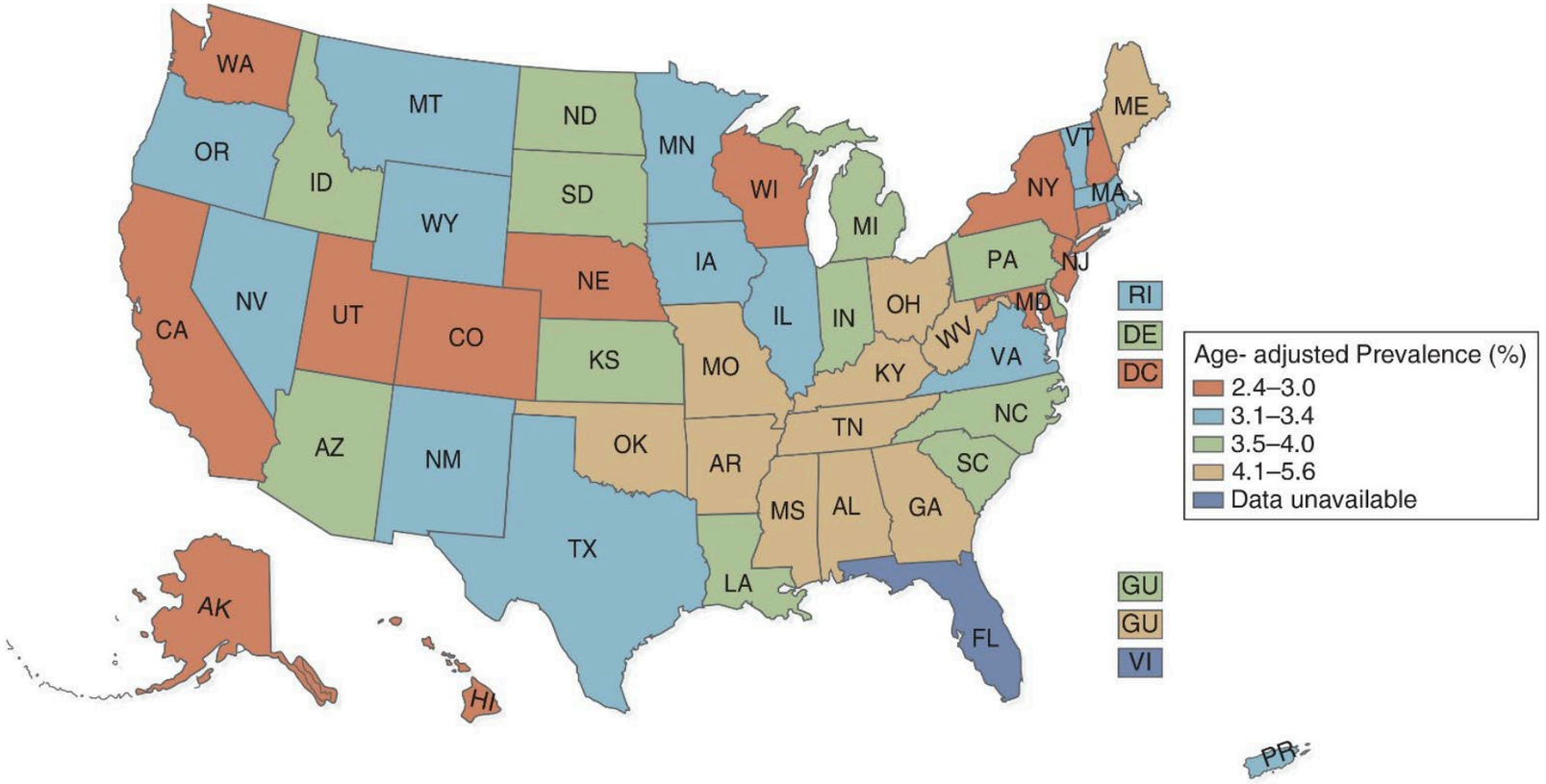


Figure 5.14 Age-adjusted prevalence of heart attack among noninstitutionalized adults age 18 years or greater, by state—Behavioral Risk Factor Surveillance System, United States, 2021

Centers for Disease Control and Prevention. Behavioral Risk Factor Surveillance System, 2021. Available at: https://www.cdc.gov/brfss/brfssprevalence/. Accessed on March 14, 2023

Question: What do you notice about the geographic pattern?

From Urban–Rural to Local

- **Definition:** Geographic health patterns can exist at **multiple spatial scales**, from broad categories to very specific locations.
- **Examples**
  - **Urban vs. rural:** Higher **motor vehicle fatality rates** in rural areas compared with urban areas in the U.S.
  - **Local areas:** Elevated **childhood lead exposure** in neighborhoods with older housing stock; higher **asthma prevalence** near major highways or industrial sites.
- **Key Point:** As place becomes more specific, patterns may point to **localized risks or exposures**.
- **Discussion Questions**
  - Do you know any examples of diseases that show localized geographic patterns?

Localized Patterns of Disease

- Associated with specific environmental conditions that may exist in a particular geographic area
- Examples:
  - Lung cancer: regions with high levels of radon gas
  - Skin lesions and increased risks of lung, bladder, and skin cancers: regions with naturally occurring arsenic in the water supply
  - Dengue fever: along the Texas–Mexico border linked to the presence of Aedes mosquitoes
  - Lyme disease: Northeast, mid-Atlantic, and upper Midwest in the U.S. (~90% of reported cases)

Time: The “When” Factor

- **Definition:** Time describes how disease occurrence and health outcomes **change or vary over time**.
- **Why Time Matters**
  - Reveals **trends, cycles, and outbreaks**
  - Helps distinguish **short-term events** from **long-term patterns**
- **Common Time Patterns**
  - **Secular trends:** Long-term changes over years or decades
  - **Seasonal or cyclic trends:** Regular patterns within a year or over cycles
  - **Point epidemics:** Sudden increases over a short period
  - **Clustering:** A higher-than-expected number of cases in a specific time period or place

Secular Trends

- **Definition:** Long-term changes in disease occurrence or health outcomes observed over **many years or decades**.
- **Why Secular Trends Matter**
  - Show whether a health problem is **increasing, decreasing, or stable** over time
  - Reflect broad changes in **population behavior, environment, or policy**
- **Examples**
  - **Smoking prevalence:** Steady decline in the U.S. over several decades
  - **Obesity prevalence:** Long-term increase across many age groups
  - **Life expectancy:** Gradual increases over time, with recent plateaus or declines in some populations
- **Key Point:** Secular trends describe **direction and magnitude of change**, not causes.
- **Discussion Question**
  - Can you think of a health outcome that has shown major changes **over the past several decades or across generations in the U.S.**?

Seasonal (Cyclic) Trends

- **Definition:** Regular increases and decreases in disease occurrence that repeat over **specific periods**, often within a year.
- **Why Seasonal Trends Matter**
  - Help anticipate **predictable changes** in disease burden
  - Support **timing of prevention and response** efforts
- **Examples**
  - **Influenza:** Higher incidence during winter months
  - **Allergic conditions:** Increased symptoms during spring and fall
  - **Foodborne illness:** Higher incidence in summer months
- **Key Point:** Seasonal trends repeat in a **predictable pattern** over time.
- **Discussion Question**
  - What health conditions tend to increase during specific seasons?

Point Epidemics

- **Definition:** A sudden increase in disease occurrence over a **short period of time**, often linked to a **common exposure**.
- **Why Point Epidemics Matter**
  - Signal a **recent or ongoing outbreak**
  - Help identify **when** exposure likely occurred
- **Examples**
  - **Food poisoning outbreak** linked to a single meal or event
  - **Legionnaires' disease** outbreak associated with a contaminated water source
  - **Chemical exposure** causing acute illness in a community
- **Key Point:** Point epidemics reflect **short-term changes**, not long-term trends.
- **Discussion Question**
  - What **time-related information** helps identify **when exposure likely occurred** during a point epidemic?

Clustering

- **Definition:** The occurrence of a **greater-than-expected number of cases** within a **specific time period**, **geographic area, or population group**.
- **Why Clustering Matters**
  - May signal **unusual aggregation** of disease
  - Helps identify patterns that warrant **closer monitoring or investigation**
- **Examples**
  - **Temporal clustering:** Increased cases of influenza during a short winter period
  - **Spatial clustering:** A higher-than-expected number of leukemia cases in a small community
  - **Spatiotemporal clustering:** Dengue cases occurring in a specific neighborhood over several weeks
- **Key Point:** Clustering describes **where and when cases group together**, not why they occur.

Comparing Time and Place Patterns

Pattern	What it represents	Key distinguishing feature	Example
Seasonal trend	Expected, recurring variation over time	Predictable and repeats regularly	Influenza peaks every winter
Temporal clustering	Descriptive observation of cases grouped in time	A higher-than-expected number of cases during a specific period; no designation implied	Unusual increase in meningitis cases over 2 weeks
Point epidemic	A sharp form of temporal clustering	Very short time window, often consistent with a single exposure	Food poisoning after one shared meal
Endemic pattern	Baseline, expected level of disease in a population	Ongoing and relatively stable over time	Malaria in parts of sub-Saharan Africa
Spatial clustering	Descriptive observation of cases grouped in place	A higher-than-expected number of cases in a specific location	Lyme disease concentrated in the Northeast U.S.
Spatiotemporal clustering	Descriptive observation of cases grouped by time and place	Unusual concentration in both dimensions	Dengue cases in one neighborhood over several weeks
Epidemic (Week 1 term)	Public health designation	Determination that observed clustering exceeds what is expected	COVID-19 epidemic in early 2020

Descriptive Epidemiology: Advantages and Limitations

Advantages

- Describes who is affected and where and when disease occurs
- Identifies patterns and disparities in populations
- Supports public health surveillance and planning
- Helps generate questions for further study

Limitations

- Cannot establish causal relationships
- Often lacks temporal order between exposure and outcome
- May be influenced by reporting or measurement differences
- Describes patterns but does not explain why they occur

Project Prep 2